

Name: \_\_\_\_\_

In the first diagram,  $\overline{PR} \perp \overline{QR}$ ,  $PQ = 2x - 1$ , and  $PR = 17$ .

- 1. Name the shortest segment from point P to line QR.
- 2. Write and solve an inequality for x.

In each diagram, the two marked angles form a right angle.  
Find x.

- 3. (figure a)
- 4. (figure b)

Answer each question and name the theorem that supports your answer.

- 5. In a plane, line a  $\perp$  line c and line b  $\perp$  line c. What can you conclude about a and b?
- 6. In a plane, line r  $\parallel$  line s and line t  $\perp$  line r. What can you conclude about t and s?
- 7. Two lines intersect to form a linear pair of congruent angles. Find the measure of each angle and describe the lines.

Use coordinates.

- 8. Find the distance from point A(−2, 4) to the line  $y = 1$ .
- 9. Segment CD has endpoints C(1, 3) and D(9, 3). Write an equation of the perpendicular bisector of  $\overline{CD}$ .

Complete the two-column proof. (Use the last diagram, where j  $\parallel$  k.)

10. Given: j  $\parallel$  k,  $\angle 1 \cong \angle 2$       Prove:  $t \perp k$

| Statements                   | Reasons  |
|------------------------------|----------|
| 1. $\angle 1 \cong \angle 2$ | 1. Given |
| 2.                           | 2.       |
| 3.                           | 3.       |
| 4.                           | 4.       |
| 5.                           | 5.       |

